

Next-gen sequencing technologies and data generation

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2024/09/04

Funding



Sponsorship



Support



A long way to the "Molecular Microbiology Era" (1980's)

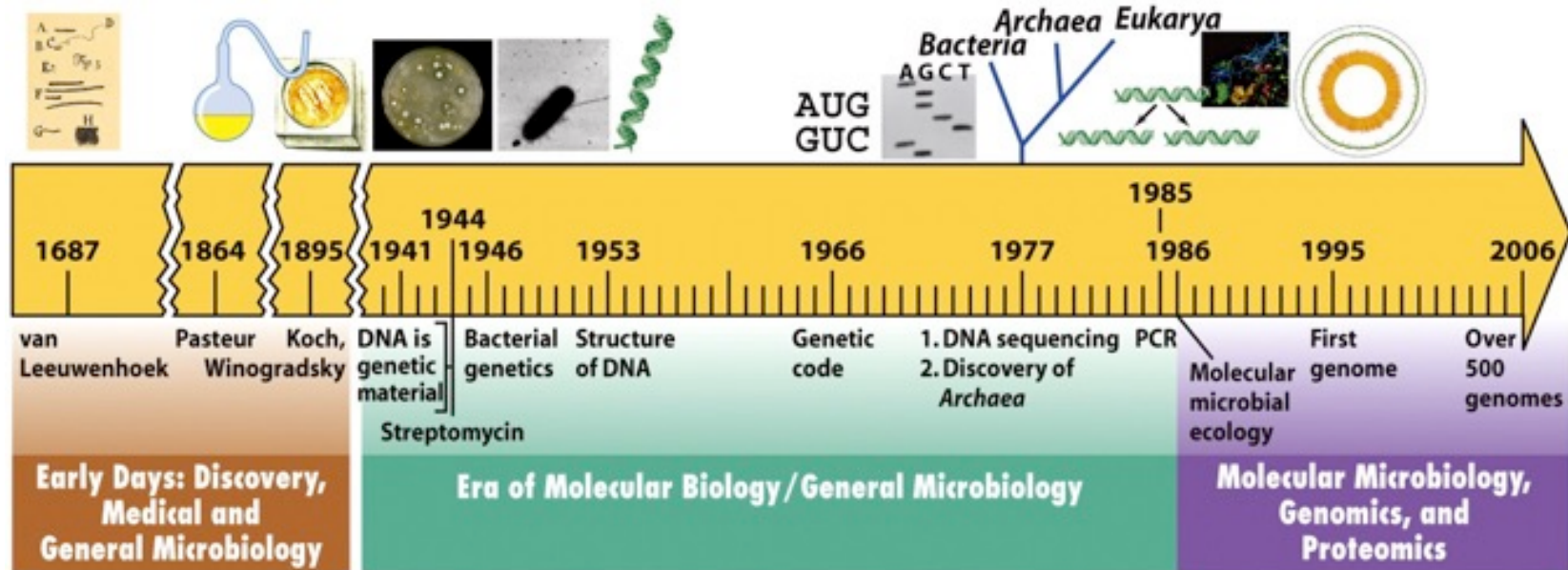


Figure 1-17 Brock Biology of Microorganisms 11/e
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- **1953 (Watson & Crick):** DNA structure (Nobel prize in Medicine and Physiology)
- **1977 (Sanger et al.):** DNA sequencing (Nobel prize in Chemistry)
- **1977 (Woese & Fox):** Ribosomal RNA analysis to recognize a third form of life (Archaea), distinct from Bacteria and Eukaryotes.

Sanger sequencing - 1977

Chain-terminating
dideoxynucleotides (ddNTPs) +
DNA Polymerase +
Electrophoresis

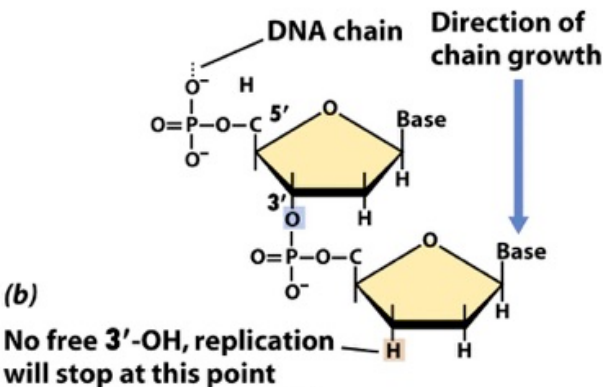
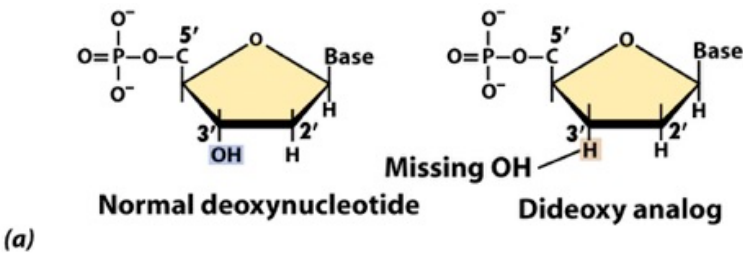


Figure 7-25 Brock Biology of Microorganisms 11/e
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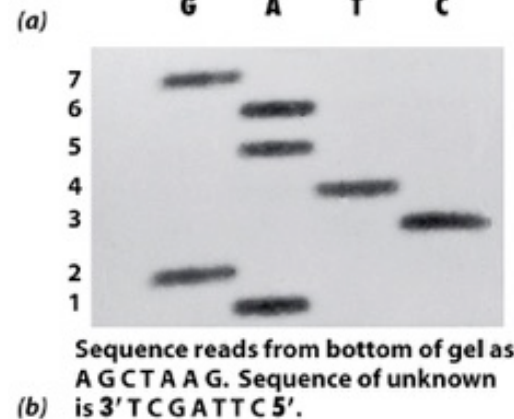
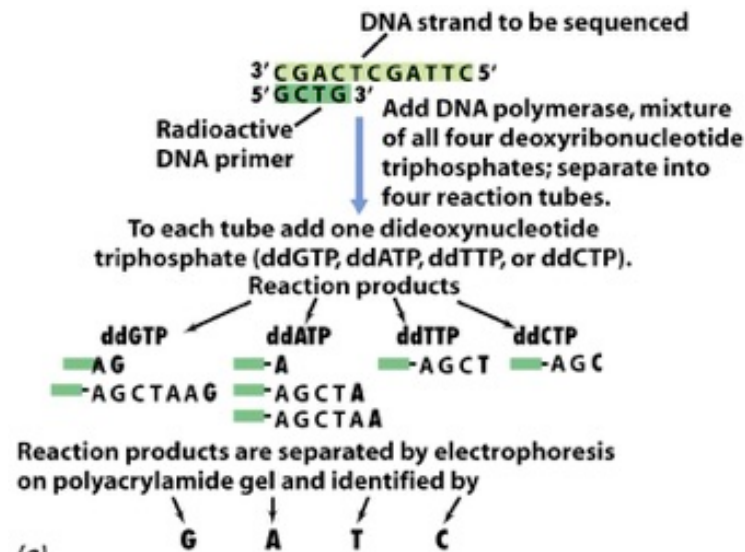
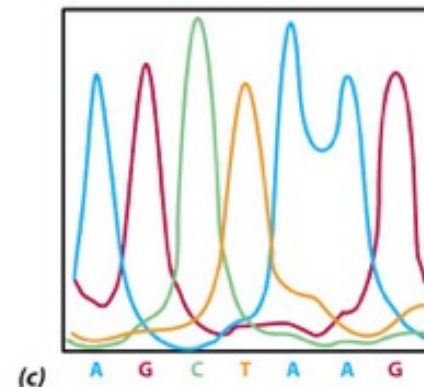


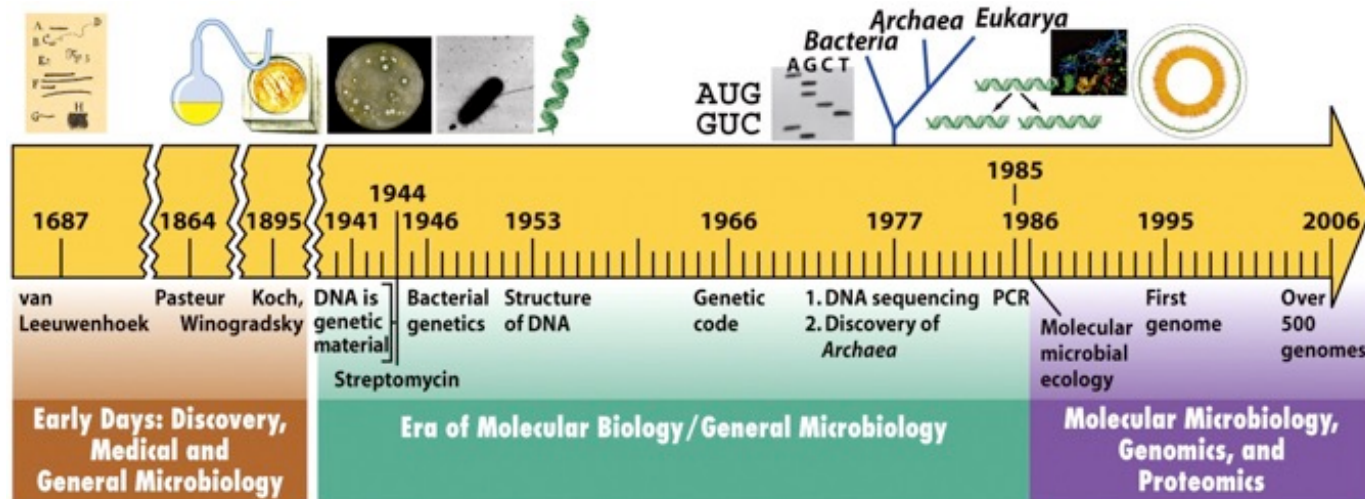
Figure 7-26 Brock Biology of Microorganisms 11/e
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Sanger sequencing – since 1977

- Read length: 400 – 900 bp
- Reads per run: ~ 400
- Total bases per run: < 360 kb
- Time per run : ~3 hrs
- Cost per 1M bases: ~\$500/Mb
- Qualities: high-quality (99.9999%), long reads
- Limitations: Slow, expensive, laborious, sequencing depth
- But, for ~30 years was virtually the only sequencing method
- **Human Genome Project (1990-2003): more than 10 years, hundreds of labs, for one species genome. What about nowadays?**

“Molecular Microbiology Era”



- **1985 (Kary Mullis):** Polymerase chain reaction - PCR (Nobel in Chemistry).
- **1986 (Norman Pace):** Investigating uncultivated microorganisms in environmental samples. DNA/RNA extraction from the environment.
- **1995 (Venter and Smith):** Completed first genome sequence of a bacteria
- **2004 (Venter et al.):** large scale of environmental genome sequences from the Sargasso Sea (shotgun sequencing)

Since then, various sequencing technologies and molecular methods were developed to determine microbial diversity in various environments

“Molecular Microbiology Era” (Next-gen sequencing)

NEXT-GENERATION SEQUENCING: high-throughput sequencing (generating hundreds of millions of sequences in a single run). Also known as “massive parallel sequencing”, “2nd generation sequencing”, “3rd generation sequencing”, NGS.

Early days (2005 – 2020)

454/Roche



Pyrosequencing

- Read length: 750 – 1000 bp
- Reads per run: ~ 1,000,000
- **Total bases per run: < 1 Gb**
- Time per run : ~24 hrs
- Cost per 1M bases: ~\$10
- long reads
- Expensive

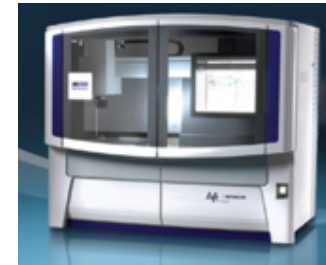
Ion Torrent PGM/Life Technologies



Semiconductor (pH change)

- Read length: 35 – 400 bp
- Reads per run: ~ 12,000,000
- **Total bases per run: < 4 Gb**
- Time per run : ~2 hrs
- Cost per 1M bases: ~\$1
- low-cost, fast
- Error rate (2%)

SOLiD/Applied Biosystems



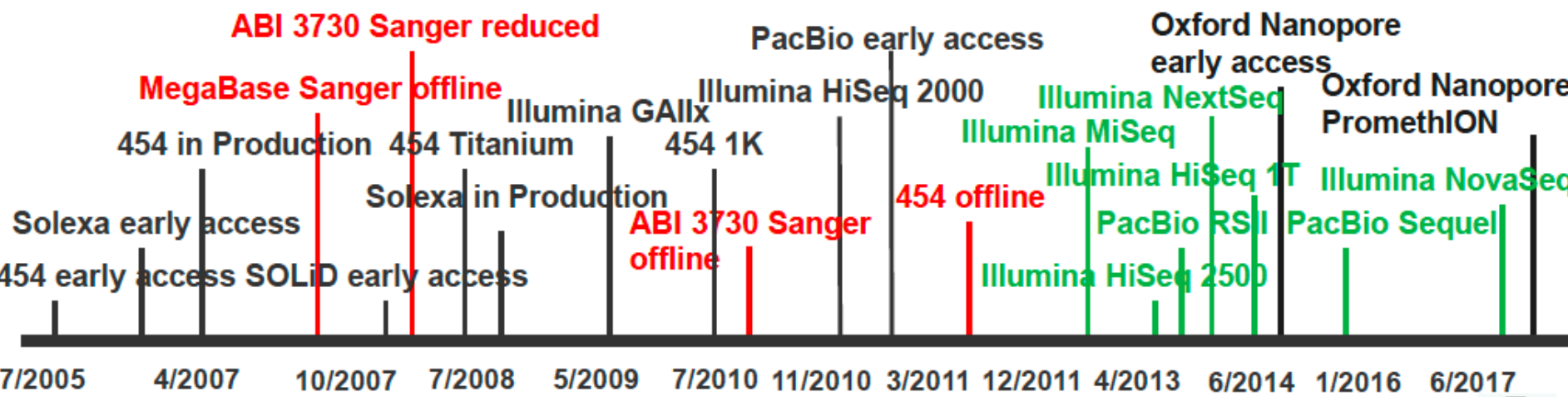
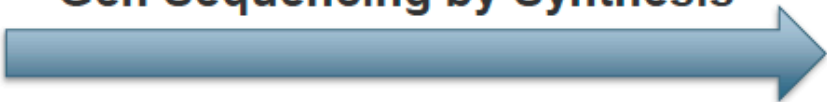
Ligation

- Read length: 75 bp
- Reads per run: ~ 400,000,000
- **Total bases per run: < 30 Gb**
- Time per run : ~10 days
- Cost per 1M bases: ~\$0.13
- accuracy (99.9%), low-cost
- Short reads

Metagenome sequencing technologies evolution (JGI)



Sanger Sequencing to Next-Gen Sequencing by Synthesis



“Molecular Microbiology Era” (Next-gen sequencing)

The current “Big-three” (since 2010)

MiSeq-HiSeq-NovaSeq/Illumina



Sequencing by Synthesis

- Read length: 2 x 250 or 150 bp
- Reads per run: 17M – 6000M
- **7 Gb (MiSeq) – 1,200 Gb (HiSeq)/run**
- Time per run : ~ 1 – 10 days
- Cost per 1M bases: ~\$0.05-0.15
- **Extremely high-throughput (most commonly used)**
- **Expensive instrument**

RSII-Sequel/PacBio



SMRTS + HiFi

- Read length: 5000 – 20,000 bp
- Reads per run: ~ 50,000
- **Total bases per run: < 1 Gb**
- Time per run : ~ 1 hr.
- Cost per 1M bases: ~\$0.10
- **No amplification needed. Long reads.**
- **Expensive. Low yield.**

PromethION-MinION/Oxford Nanopore



Minion



MinION MkI: portable, real time biological analyses

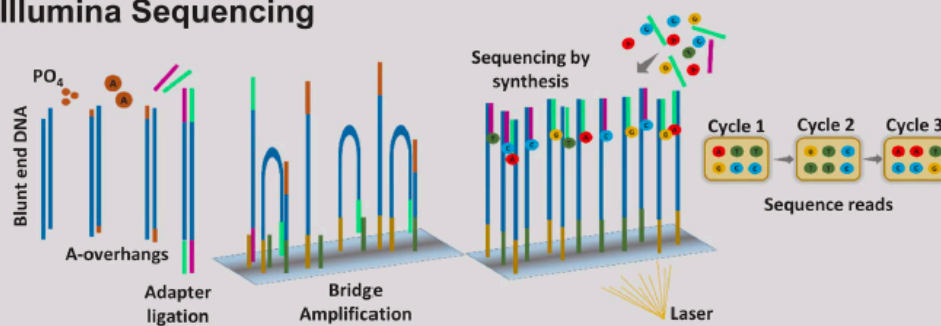
MinION

Current change through pores

- Read length: 300,000 – 4M bp
- Reads per run: ~ 1M
- **Total bases per run: 3 Gb – 15 Tb**
- Time per run : ~48 hrs
- Cost per 1M bases: \$0.03\$
- **Long reads, portable. Affordable. No amplification needed**
- **Low accuracy**

Metagenome sequencing technologies

A. Illumina Sequencing

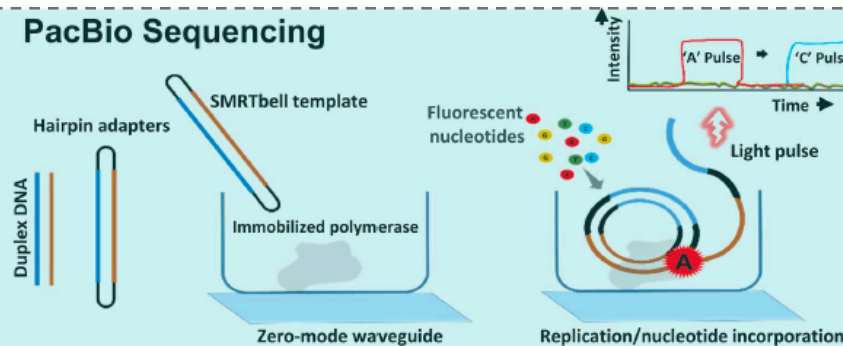


Short reads (sequencing depth)

Single or Paired reads generated (e.g. 2 x 150). Useful overlap.

Amplification needed (bias)

B. PacBio Sequencing

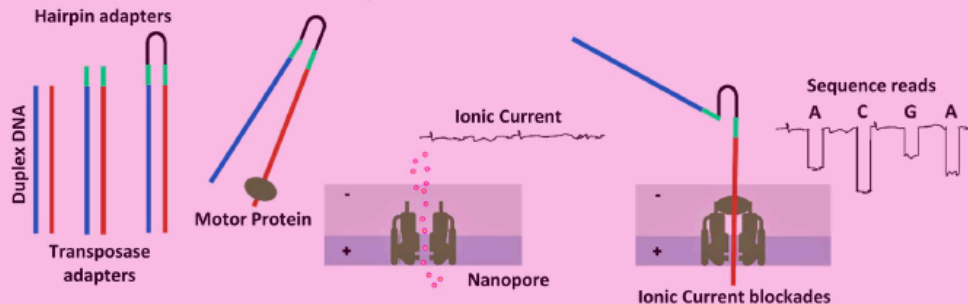


Long reads (sequencing length)

No amplification needed

Sample quality limitation

C. Nanopore Sequencing



Long reads (sequencing length)

No amplification needed

Sample quality limitation

Generated Data

Main output from any sequencing technology is a **FASTQ file**

- Text files that store biological information in the form of strings of nucleotides
- Formally defined in 2010
- Gold standard file format for storing and representing biological sequence data

@ and sequence identifier	—	@M04577:12:000000000-D2N7B:1:1101:15525:1827 1:N:0:23
raw sequence	—	TCTTTCGCTGGCGCGTAACAGCAAAAGCAGGTACTGATTCAAATTGTTTGCTTTGTTCTTCTTTGCTTCATTCCTTTTTTCTTCTTTC
		TTTCTTTTTTTCTTTTTTTTTTTTTTTTTTTTCTTTTATTCTTTTCTTTCTTTTTTCTCTTTTTTTTTCTCTCTTTTTTTTTCTTTTTCTT
		TTTTTCTTTTTTCTTTTCTTCTTTTTTTTTTTTTTTTCTTTTTTTTTTTTTTCTTTTTTCTTTTT
metadata	—	+
Quality score per base call	—	1AAAAFAAAADAGGGGGEBBC00100010/B1AB22DD21222B1220002DDB1B0121221A11DD112D222B11>01@22112BB1221111///0
		11B11/////>/>-----/000000:000000:000000:9/00:0000:-9-///:::9/----9/:/9B//:::/9///:9@///::B//:://-----9----9//9//:9-9----:/:/-9/////:

What is a quality core?

$$Q_{\text{score}} = -10 \log_{10} p$$

p = probability that some base was called incorrectly

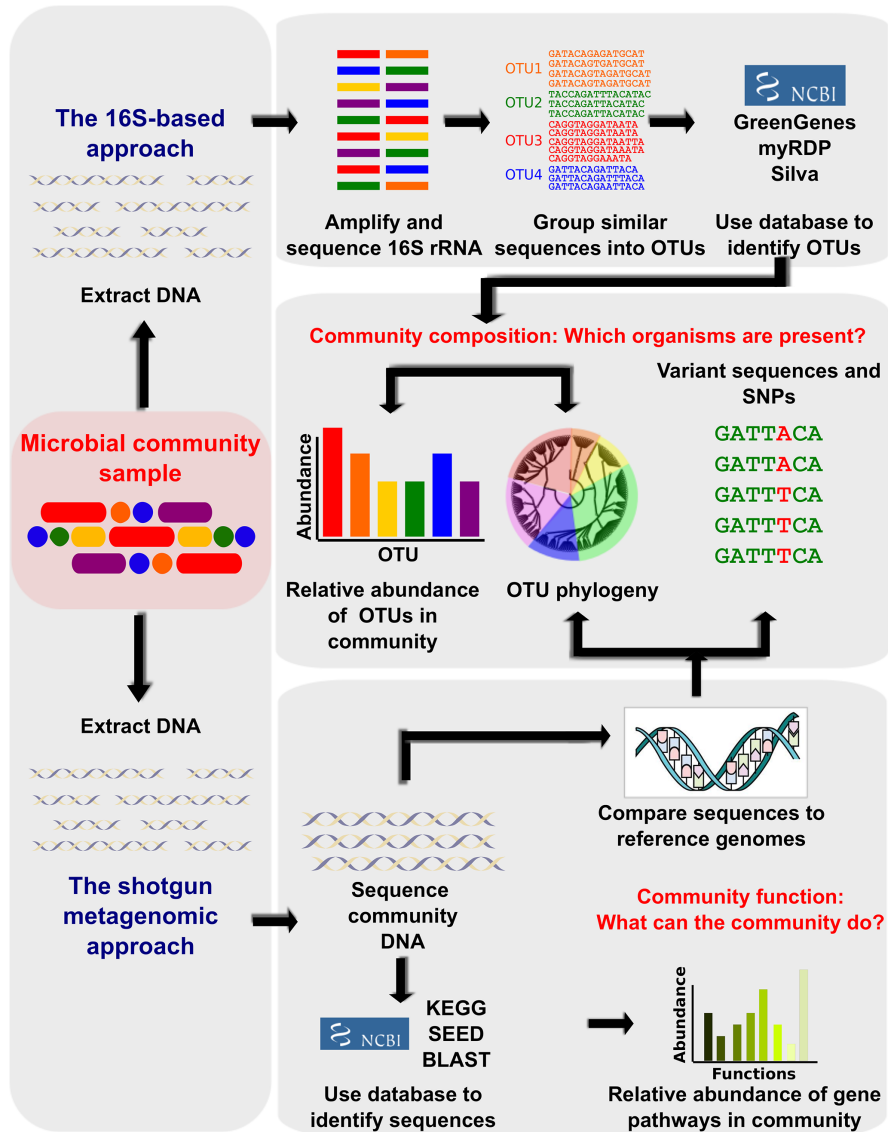
Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10,000	99.99%
50	1 in 100,000	99.999%
60	1 in 1,000,000	99.9999%

Table source: wikipedia.org

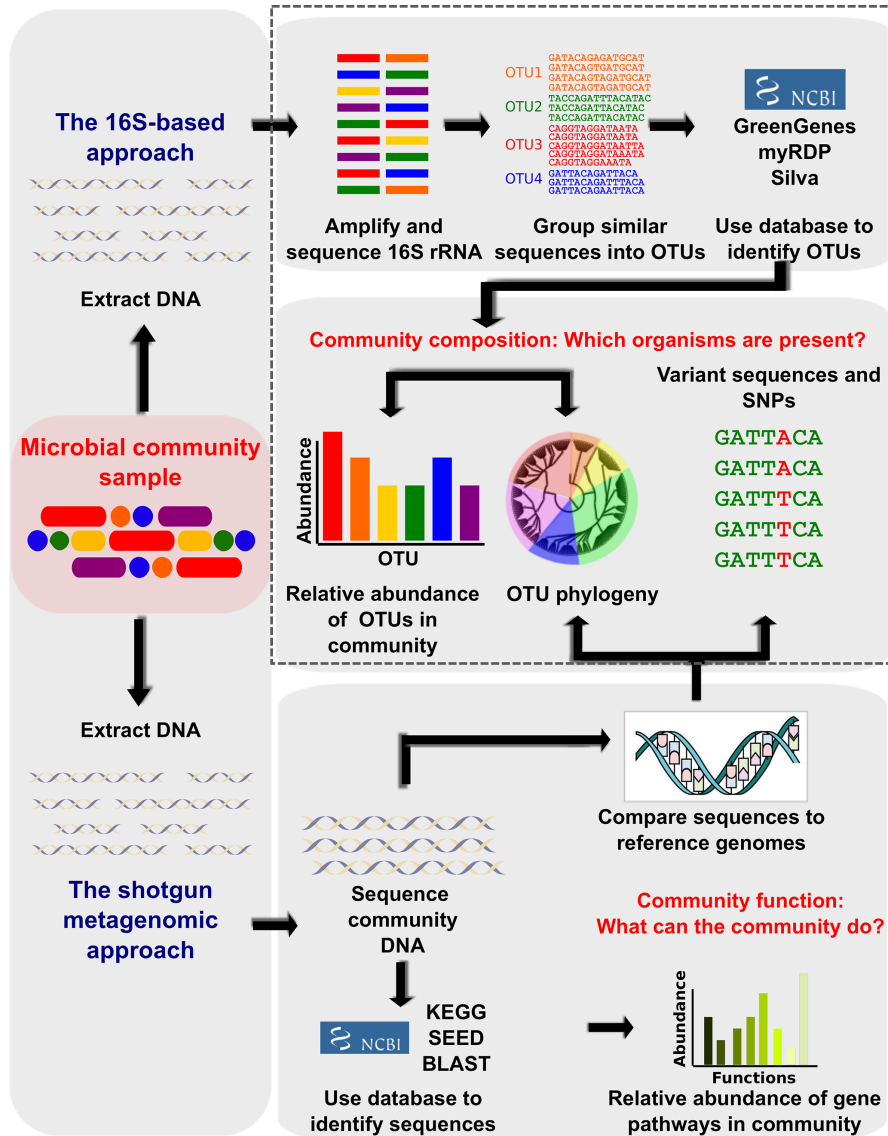
General Applications of NGS

- Genome sequencing
- RNA-Seq (gene expression, exon-intron structure)
- CHIP-Seq (protein-DNA interactions)
- Single nucleotide polymorphisms
- Epigenetics
- Metagenomics: “*genomic analysis of microorganisms by direct extraction ~~and cloning~~ of DNA from an assemblage of microorganisms*” (Handelsman et al., 2004). Aka as environmental and community genomics.

Targeted and Untargeted Metagenomics



Targeted and Untargeted Metagenomics

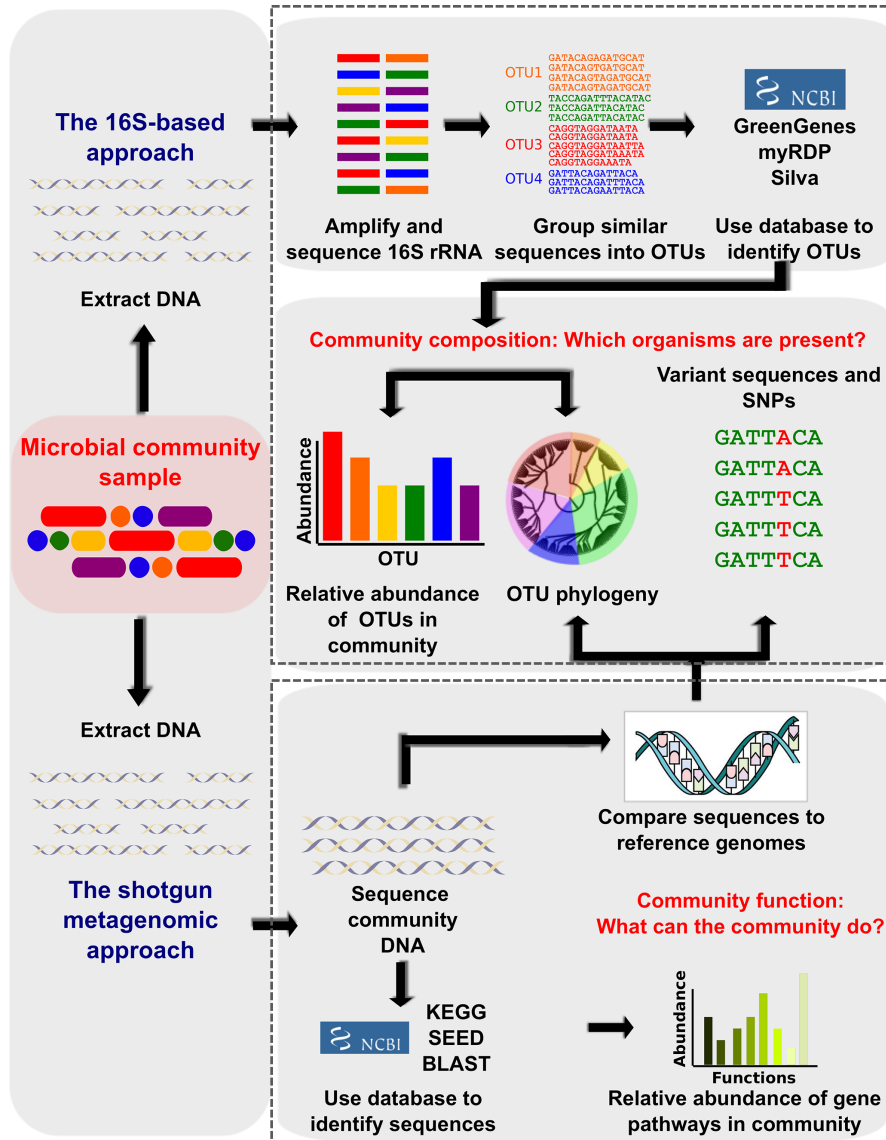


Targeted Metagenomics / Metabarcoding / Amplicon Sequencing

Marker genes amplification (e.g. 16S rRNA, others)

- Who's there (in general)?
 - With taxonomic markers: What can they possibly do (functional inference)?
 - With functional genes: Who's doing this specific function (potentially)?

Targeted and Untargeted Metagenomics



Targeted Metagenomics / Metabarcoding / Amplicon Sequencing

Marker genes amplification (e.g. 16S rRNA, others)

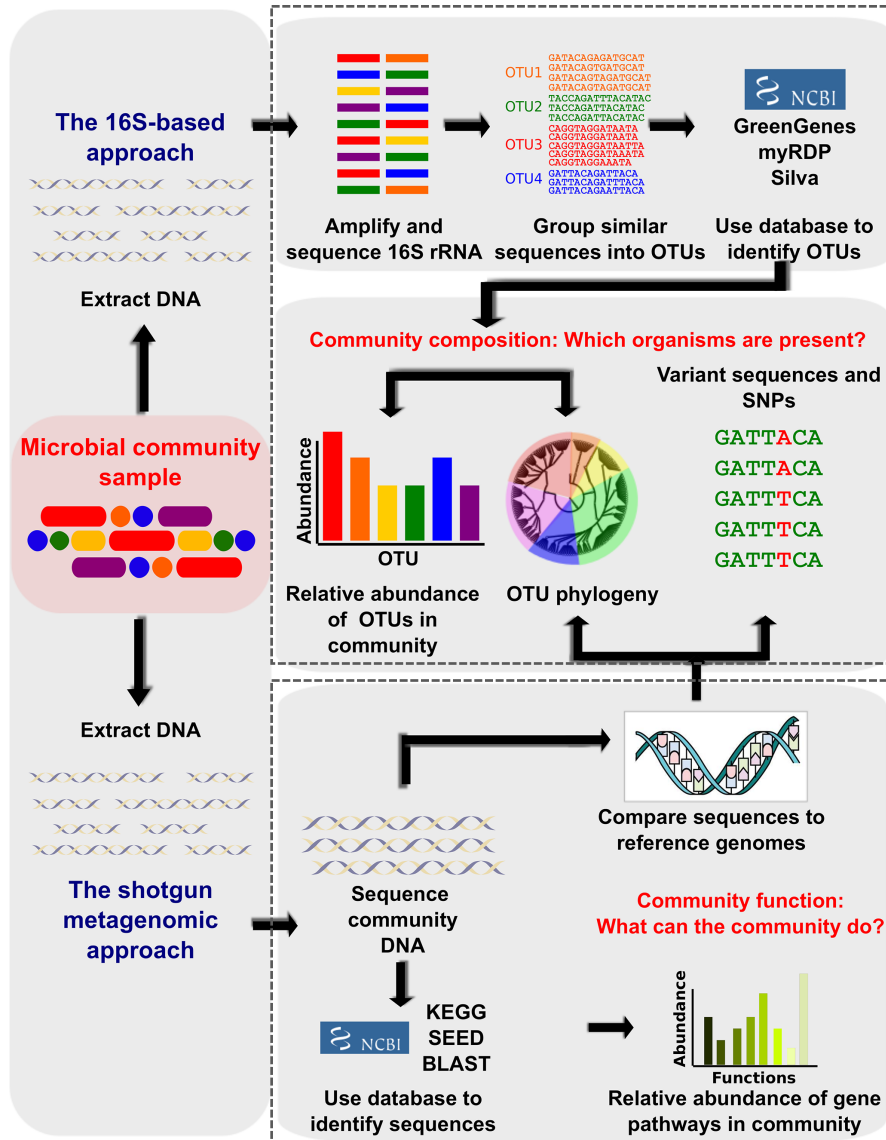
- **Who's there (in general)?**
 - With taxonomic markers: **What can they possibly do (functional inference)?**
 - With functional genes: **Who's doing this specific function (potentially)?**

Untargeted Metagenomics / Shotgun Sequencing / Whole Genome Sequencing

All gene fragments (no PCR amplification)

- **Who's there (in general)?**
- **What can they do (potentially)?** Across infinite functions. **FUNCTIONS UNCHAINED!**

Targeted and Untargeted Metagenomics



Targeted Metagenomics / Metabarcoding / Amplicon Sequencing

MAIN LIMITATIONS:

- PCR-biased.
- Functional discovery limited to taxonomic classification (for 16S).

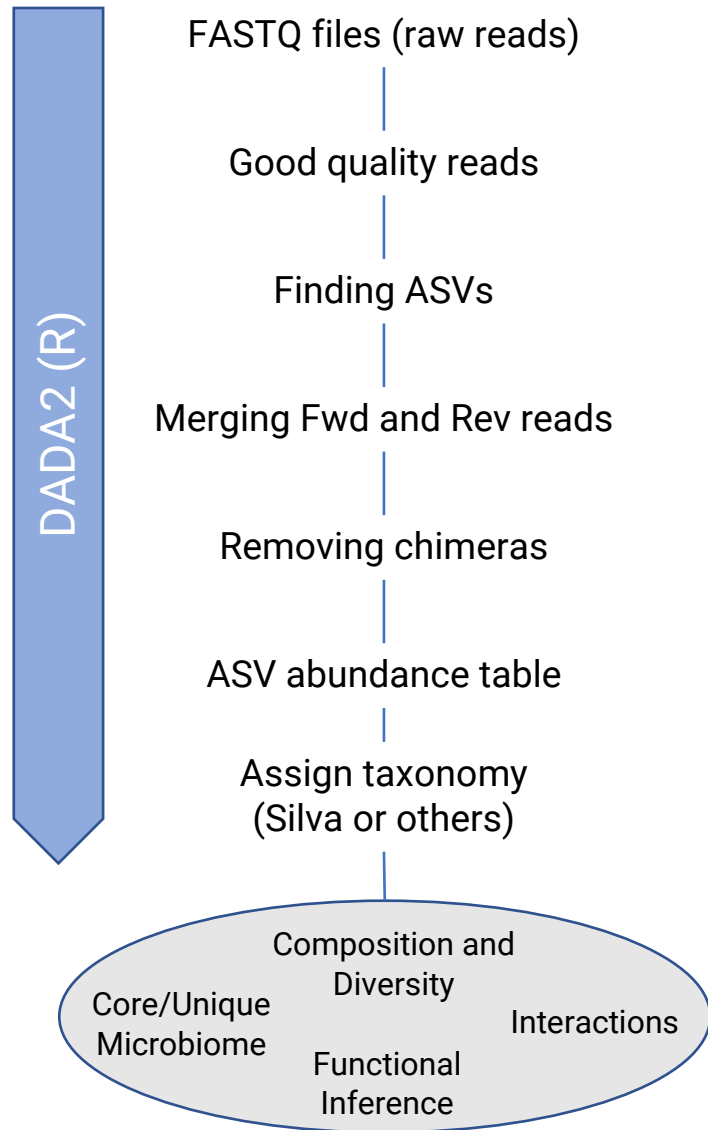
Untargeted Metagenomics / Shotgun Sequencing / Whole Genome Sequencing

MAIN LIMITATIONS:

- Sequencing depth.
- Cost (€).

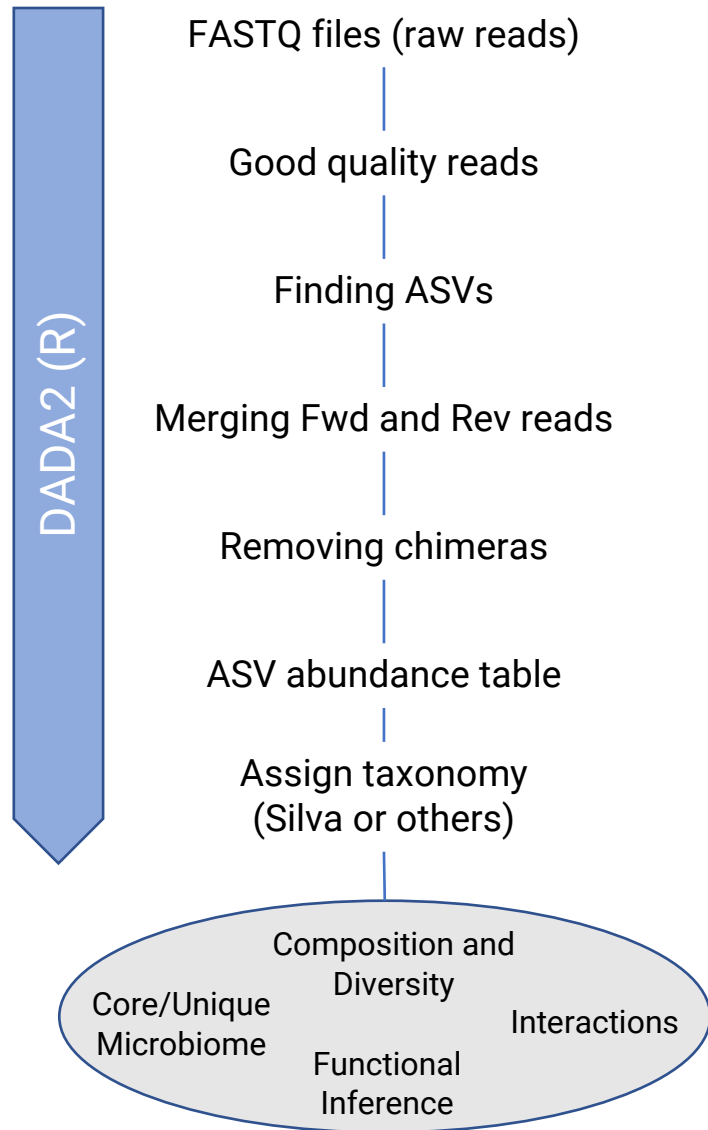
Targeted and Untargeted Metagenomics

Amplicon Sequencing (e.g. 16S rRNA)

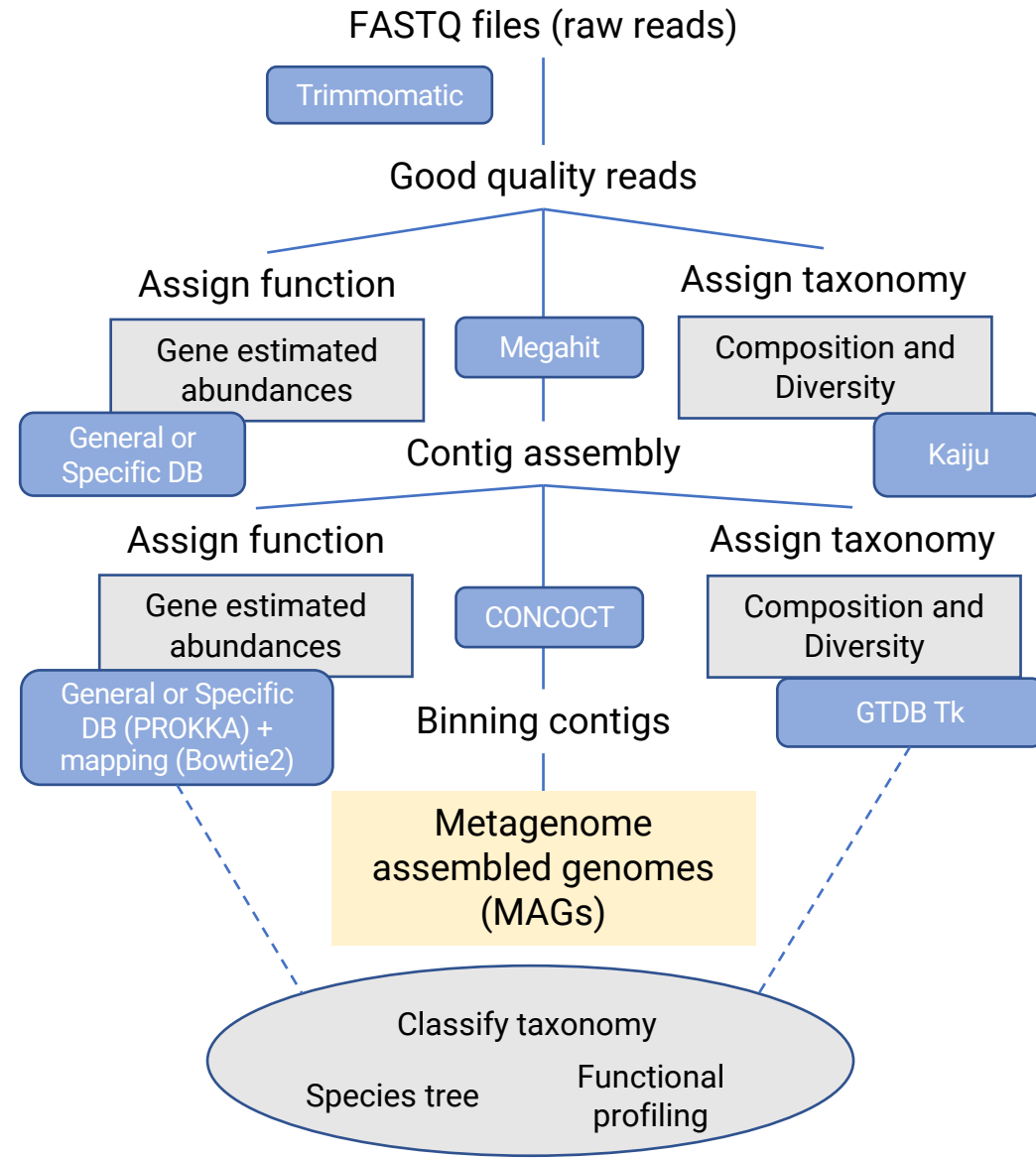


Targeted and Untargeted Metagenomics

Amplicon Sequencing (e.g. 16S rRNA)

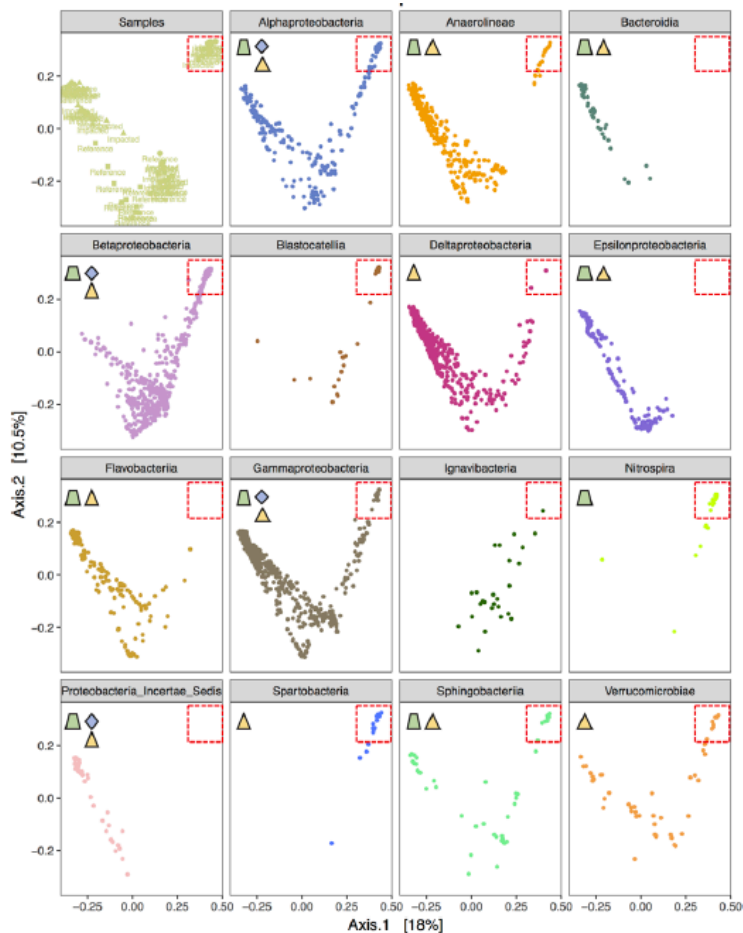


Whole Genome Sequencing

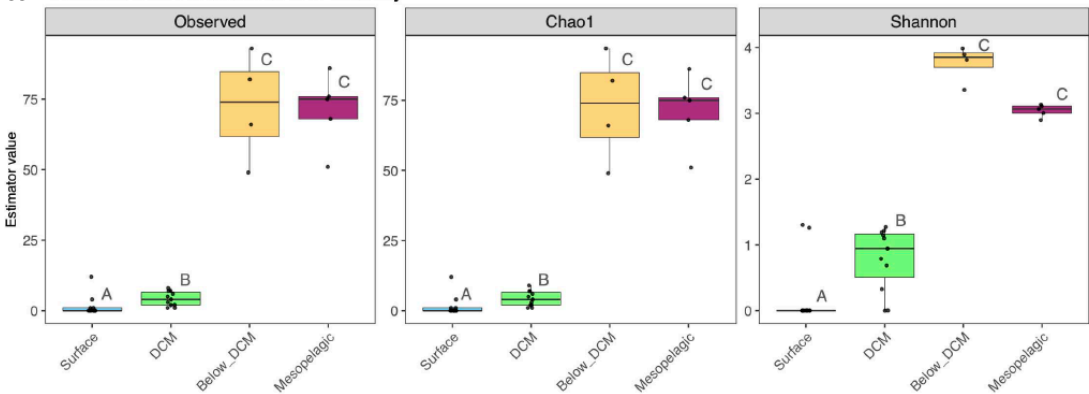


What to do with amplicon-based sequences

Composition and Diversity

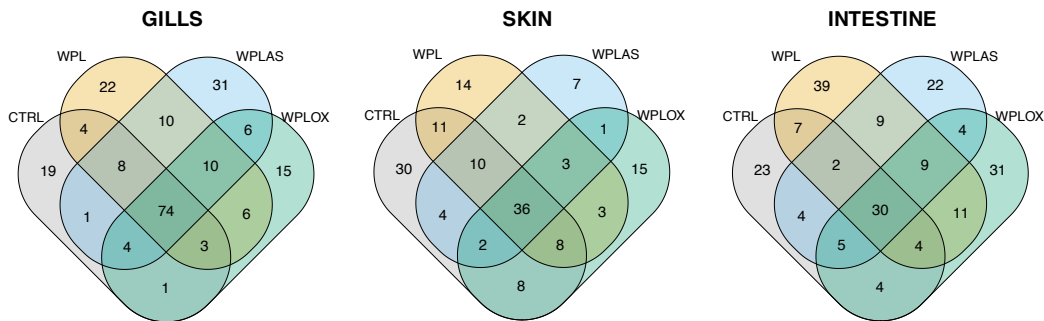


A Thaumarchaeota richness and diversity



Semedo et al, 2021. Front. Microbiol.

Core/Unique Microbiome

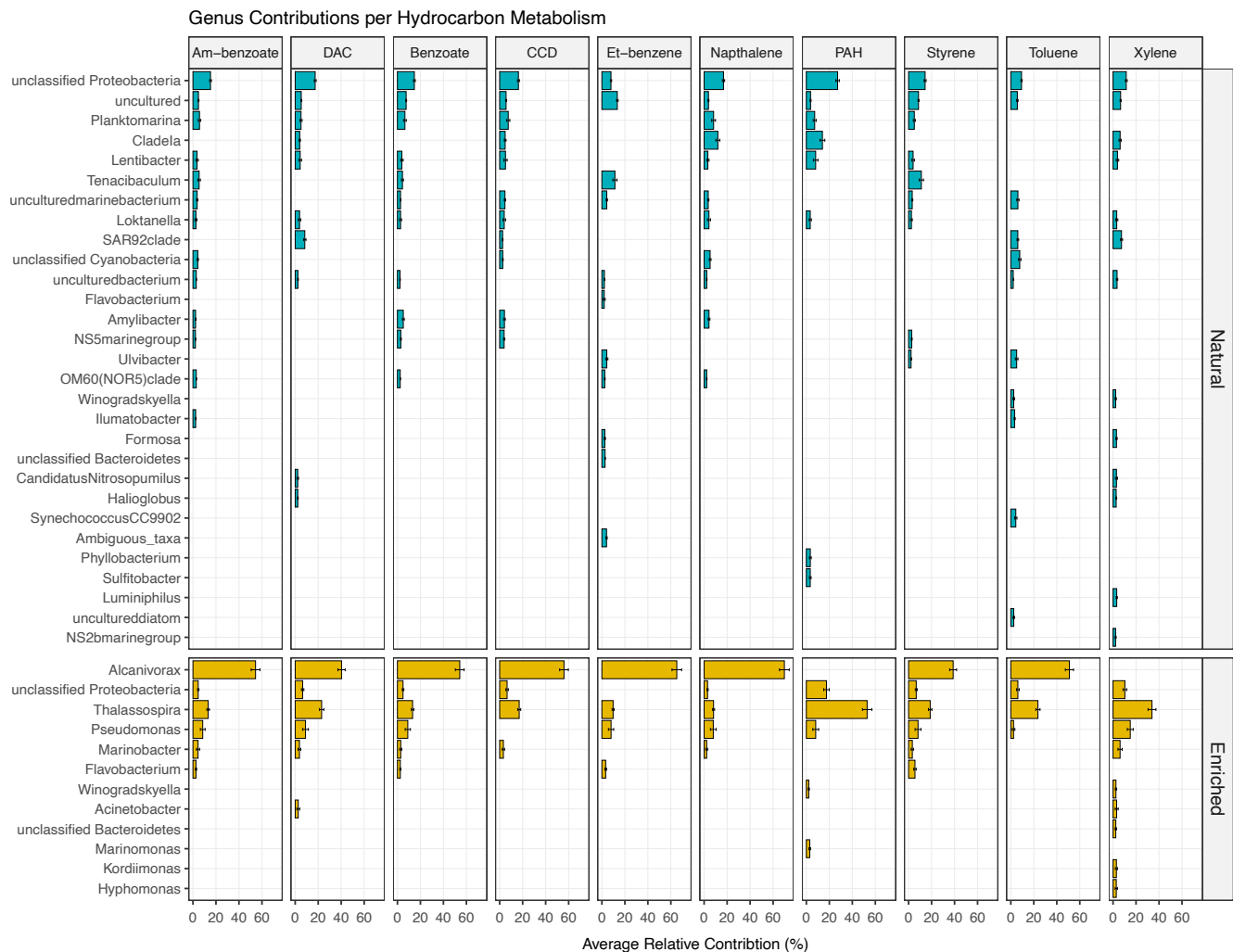


Semedo et al., 2024, in prep

Denitrification genes: $\triangle$ nirS $\diamond$ nosZ-I $\triangle$ nosZ-II

What to do with amplicon-based sequences

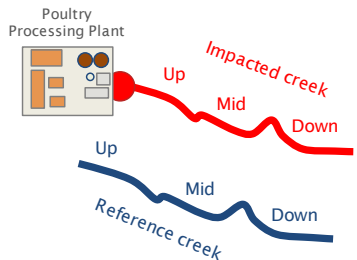
Functional inference



Cautiously

Gene estimated abundances

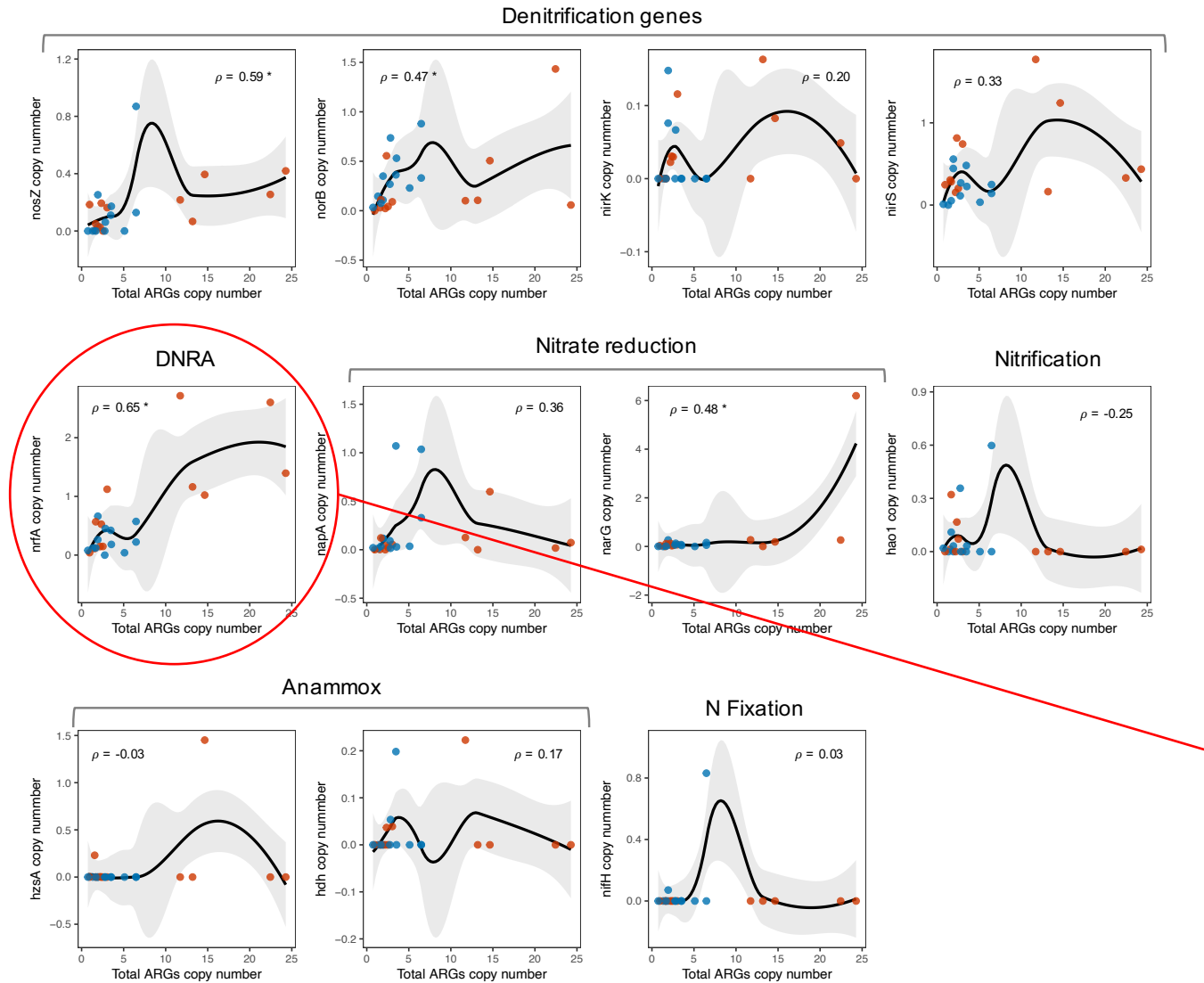
■ Impacted ■ Reference



What to do with metagenome sequences

Gene estimated abundances

Gene or metabolic pathway relationships



What to do with metagenome sequences

At the individual genome level!

Pathway co-occurrence

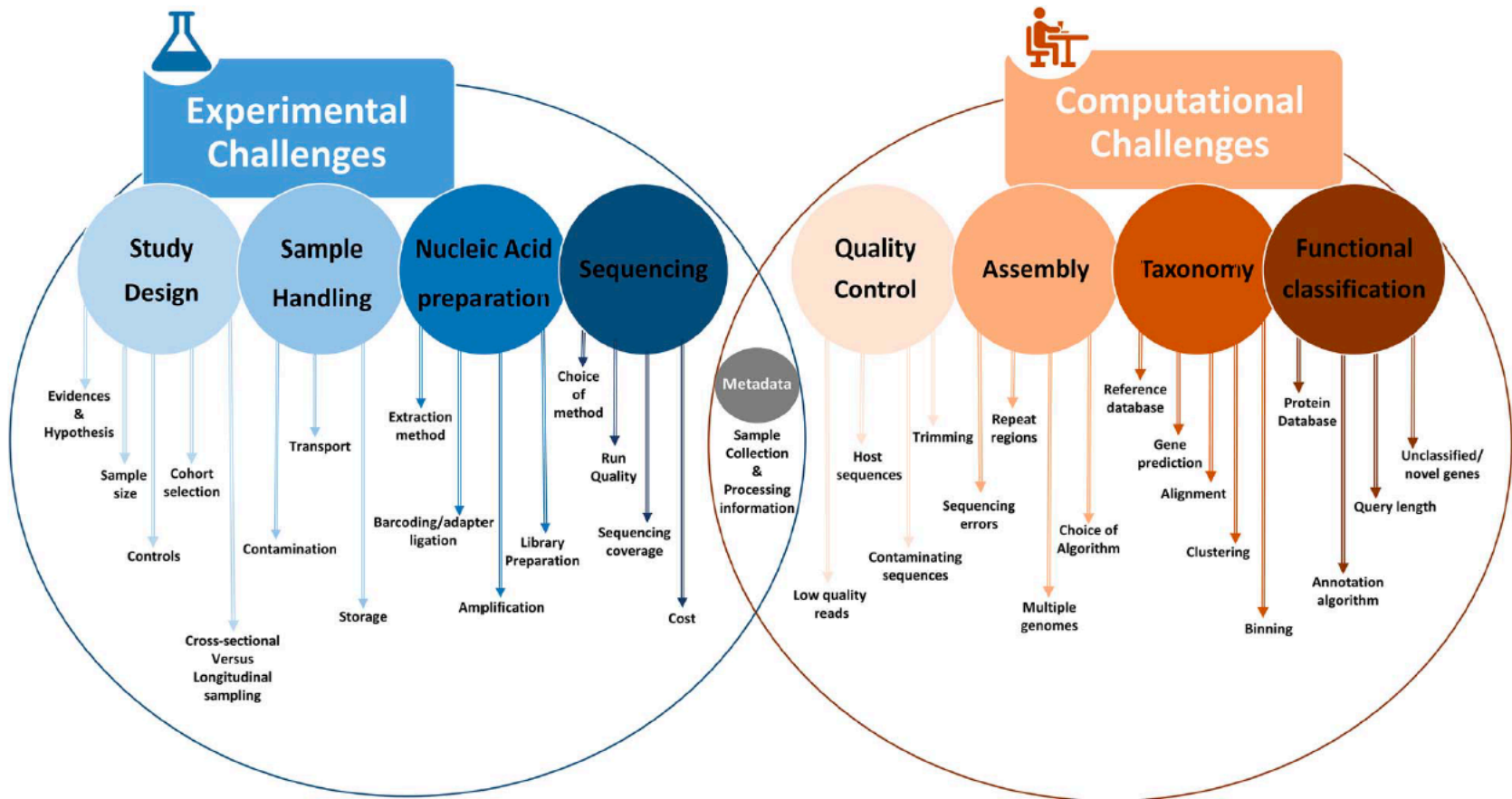
Taxonomic identification

Functional profiling

Metagenome assembled genomes (MAGs)

Station	Creek	MAG ID	Genus (class/phylum)	# ARG	Nitrate red.		DNRA	Denitrification				Nitrif.	Anammox		N fix.	% comp.	% cont.
					narG	napA	nrfA	nosZ	norB	nirK	nirS	hao1	hzsA	hdh	nifH		
Upstream	Impacted	bin.010	Piscinibacter (Gamma-proteob.)	6		yes	yes				yes					95,6	5,7
		bin.013	UBA3961 (Bacteroidia)	6				yes								87,1	2,0
		bin.057	unclass. (Verrucomicrobiae)	6		yes										99,3	7,4
		bin.017	PHOS-HE28 (Bacteroidia)	5				yes								95,1	3,6
		bin.018	Aestuiriivirga (Alpha-proteobacteria)	5	yes	yes	yes			yes						93,2	4,2
		bin.025	unclass. (Acidimicrobiia)	5												95,9	1,8
		bin.037	JAAZBK01 (Acidimicrobiia)	5		yes										94,0	6,3
		bin.042	JJ008 (Bacteroidia)	4				yes								77,6	1,1
		bin.055	JACADZ01 (Alpha-proteobacteria)	4	yes		yes									94,4	2,1
		bin.061	unclass. (Thermoanaerobaculia)	4	yes	yes										66,9	8,4
		bin.016	Methylomirabilis (Methylomirabilia)	3	yes	yes					yes			yes		90,2	3,5
		bin.038	UBA7227 (Anaerolineae)	3	yes	yes	yes				yes					70,4	4,2
		bin.046	JAAUPO01 (Cyanobacteriia)	3												84,1	1,0
		bin.047	unclass. (Kapabacteria)	3		yes		yes								86,9	0,1
		bin.050	Terricaulis (Alpha-proteobacteria)	3	yes											81,6	4,9
		bin.051	unclass. (Ignavibacteria)	3		yes	yes									86,5	1,9
		bin.056	unclass. (Myxococcota)	3	yes	yes					yes					60,7	4,4
		bin.006	unclass. (Gamma-proteob.)	2		yes		yes			yes					95,7	1,9
		bin.019	UBA8403 (Bacteoidia)	2												68,9	1,0
Ref.		bin.009	Sulfuricella (Gamma-proteob.)	2		yes	yes	yes	yes		yes					85,7	6,9
		bin.004	SM1-50 (Thermoplasmatota)	0												93,2	3,2
		bin.013	PALSA-986 (Thermoproteota)	0												55,5	1,9

Challenges associated with NGS-based research



Questions?